Proceedings of Machine Learning Research vol vvv:1–15, 2027

One Budget for Every Gap: Certifying Reduced-Order, Quantized, Sampled-Data Neural Barrier Controllers on the Full Plant

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Abstract

A neural safety filter for a high-dimensional plant is verified on a reduced model, in exact arithmetic and continuous time, but runs as a quantized network and a look-up table, sampled and latched, on the full plant under a bounded disturbance. We charge each gap to one term of a single control-authority budget on a reduced control barrier function. The reduced-order tail, the table quantization, the verified deviation of the quantized network, the latch lag, the sample-and-hold error and the adversary each consume part of the certified authority δ_{cert}. When their sum does not exceed it, and under standing hypotheses on the plant, the reduction and the initial state, the full-order plant is safe for all time. For underdamped second-order port-Hamiltonian plants with uniform modal damping and $m \leq K$ actuators, δ_{cert} has a closed-form lower bound. A coupled twin-network CROWN pass bounds the deviation of the quantized network from its teacher, and both this bound and a closed-form companion are of order 2^{-b_B}. On a 320-state reaction–diffusion–advection lattice at Péclet 20, open-loop unsafe on every seed, the budget closes at utilisation 0.931 and no certified trajectory leaves the safe set. Without the tail or quantization term, the certificate accepts a controller the plant then violates, and the single-rate alternative is refused at the deployed period. The certified plant filters its advection spectrally, and a sweep over the filter weight maps where the L^2-based tail bound is non-vacuous; unfiltered advection is refused. Every constant is computed in double precision from the operators, and the code reproduces every number.

Keywords: safety-critical control; control barrier functions; neural network verification; formal verification of learning-based controllers; sampled-data control; reduced-order modeling; quantized neural networks; port-Hamiltonian systems; physics-constrained systems; robust safety certificates

1. Introduction

A neural safety filter is rarely verified on the system it runs on. It is designed on a reduced model because the plant is too large. It is specified as a floating-point teacher, although the deployed student is a b_B-bit network paired with a b_A-bit look-up table. It is analysed in continuous time because that is where barrier conditions are stated, while the hardware samples the state at period T_s and latches two commands after two different delays. It is certified against a nominal model, while the plant is driven by an unknown bounded disturbance. Each of these gaps has its own remedy; what is missing is a statement of what survives all of them at once, on the plant, for all time.

We give one: each of the six gaps is charged to one term of one inequality. When the inequality closes, under standing hypotheses on the plant, the reduction and the initial state, the full-order plant is safe for all time; when the tail or the quantization term is dropped, the certificate accepts a controller the plant then violates. The single-rate alternative, refused at the deployed period, leaves the safe set at a longer one, and so does the plant under an adversary beyond the certified level. Every gap consumes part of the certified control authority δ_{cert} of a reduced barrier, the amount by

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which the teacher holds the barrier rate above its prescribed decay on a shell near the boundary. The terms are not safety factors stacked on a nominal design; each bounds one summand of the exponentially weighted integral of the reduced barrier rate over one sampling interval along the full-order flow.

On a 320-state telegraph reaction–diffusion–advection lattice at Péclet 20, open-loop unsafe on all 16 seeds, the deployed configuration consumes 0.931 of its certified authority and no certified trajectory leaves the safe set. The certified plant filters its advection spectrally, unfiltered advection is refused, and a sweep over the filter weight locates where the certificate stops accepting. The contributions are four. (i) The budget (7), named the Certified Control Authority Condition, and Theorem 1: full-order safety in continuous time, for all time and under a bounded adversary, of a reduced-order, quantized, multi-rate sampled-data neural controller (Sections 3 and 4). (ii) A closed-form lower bound on δ_{cert} for class-(H) reduced models with uniform modal damping, $\gamma < 2\omega_{\min}$ and $m \leq K$ actuators, within one percent of brute force in both regimes (Lemma 1, Table 1). (iii) A bound on the deviation of a quantized GroupSort-2 student from its isometric teacher, obtained by a twin-network CROWN pass, an instance of differential verification, with a closed-form companion; both are of order 2^{-b_B} (Proposition 1, Section 5). (iv) The certificate on a 320-state lattice in two regimes, with ablations, refusals and the frontier sweep (Section 5, Figure 1). Section 2 fixes the setting, Section 3 derives the six terms, Section 4 states the theorem, Section 5 certifies the lattice and Section 6 states what the certificate does not cover.

Sampled-data barriers. Control barrier functions certify forward invariance for continuous-time control-affine systems (Ames et al., 2017, 2019). Sampled-data versions tighten the condition to hold between samples, under input delay (Singletary et al., 2020), for a fixed time step (Breeden et al., 2022), through approximate discrete-time models (Taylor et al., 2022), or by a zero-order condition free of derivatives (Tan et al., 2025). The closest work is Niu et al. (2024), which bounds the gap between digital and true barrier constraints under sampled, quantized state measurements and finite-precision coefficients on the full-order plant; here the quantized object is a learned controller and a reduced-order tail term is added. These works charge neither the tail term nor the verification slack.

Reduced-order and model-error barriers. Robust and input-to-state safe barriers add a margin for an unstructured disturbance (Jankovic, 2018; Kolathaya and Ames, 2019). Reduced-order designs filter on a low-dimensional model and carry a tracking error to the full system (Singletary et al., 2022; Molnar et al., 2022; Cohen et al., 2024). The closest reduced-order work is Cohen et al. (2025), which relates the two models by projection maps, bounds their discrepancy by an assumed supremum and proves invariance of a safe set inflated by that bound. Here the residual is computed from the Petrov–Galerkin truncation of a dissipative port-Hamiltonian operator and enters as one budget term; these works charge no quantization, slack, lag or hold term.

Verified and quantized networks. Bound propagation gives sound linear bounds on a network's output over an input box (Zhang et al., 2018; Xu et al., 2020; Wang et al., 2021), GroupSort networks carry a provable Lipschitz constant (Anil et al., 2019), and semidefinite programs bound that of a general network (Fazlyab et al., 2019), though Pauli et al. (2022) showed the coupled-neuron variant invalid. Differential verification bounds the gap between a network and its compressed or quantized copy (Paulsen et al., 2020; Zhang et al., 2023b). Neural barriers and Lyapunov controllers are

verified, exactly or by CROWN-type bounds, on a fixed model with real-valued weights (Zhang et al., 2023a; Yang et al., 2024; Dawson et al., 2023); none charges the tail, lag or hold term.

Structure-preserving reduction. Port-Hamiltonian and GENERIC models encode storage and dissipation in their structure (van der Schaft and Jeltsema, 2014; Grmela and Öttinger, 1997). Structure-preserving reduction by interpolation (Gugercin et al., 2012), POD/DEIM (Chaturantabut et al., 2016) or discretization of boundary-controlled PDEs (Toledo-Zucco et al., 2024) keeps the reduced model passive; matrix measures and non-Euclidean contraction bound error growth (Söderlind, 2006; Davydov et al., 2022). Califano (2023) pairs barrier filters with port-Hamiltonian structure on the full-order model.

Stochastic and sector-bounded analyses. Stochastic barrier certificates bound the probability of reaching the unsafe set (Prajna et al., 2007; Kushner, 1967; Santoyo et al., 2021; Jagtap et al., 2021; Cosner et al., 2024), the probability-one guarantee of the zeroing barriers of Clark (2021) was shown not to hold by So et al. (2023), and classical quantized feedback treats quantization as a stability problem (Delchamps, 1990; Elia and Mitter, 2001; Fu and Xie, 2005); none of them charges the tail term, the verification slack or the lag term.

2. Setting

Plant and operating domain. The plant is $\dot{x} = F(x) + Gu + Dw$ on $\mathbb{R}^N$, with state in a compact operational design domain $\mathcal{X}_{\mathrm{ODD}}$ containing the origin in its interior, input $u \in U_{\mathrm{adm}} := \{\|u\|_2 \le u_{\max}\}$, $u \in \mathbb{R}^m$, and measurable disturbance $w(t) \in \mathcal{B}(\varepsilon_{\mathrm{adv}}) := \{\|w\|_2 \le \varepsilon_{\mathrm{adv}}\}$; solutions are Carathéodory. Let ρ_{ODD} be the radius of the largest origin-centred ball in $\mathcal{X}_{\mathrm{ODD}}$.

Assumption 1 (Plant) *$F(x) = A_{\mathrm{full}}x + F_{\mathrm{nl}}(x)$ with $A_{\mathrm{full}} := (J_{\mathrm{full}} - R_{\mathrm{full}})P_{\mathrm{full}}$, $J_{\mathrm{full}} = -J_{\mathrm{full}}^{\top}$, $R_{\mathrm{full}} = R_{\mathrm{full}}^{\top} \succeq 0$, $P_{\mathrm{full}} \succ 0$, $F_{\mathrm{nl}}(0) = 0$, F_{nl} locally Lipschitz on $\mathbb{R}^N$ and $L_{F_{\mathrm{nl}}}$-Lipschitz on $\mathcal{X}_{\mathrm{ODD}}$.*

Two classes are treated. In class (D), $R_{\mathrm{full}} \succ 0$ and $P_{\mathrm{full},\epsilon} := P_{\mathrm{full}}$. In class (H), the plant is an underdamped second-order Hamiltonian system, $x = (q, p)$, J_{full} canonical, $P_{\mathrm{full}} = \mathrm{blkdiag}(\Omega_{\mathrm{full}}^2, I)$, $R_{\mathrm{full}} = \mathrm{blkdiag}(0, R_p)$, and for a damping level $0 < \gamma < 2\omega_{\min}$ (with $\omega_{\min}$ the smallest full-order natural frequency) the Lyapunov metric is $P_{\mathrm{full},\epsilon} := \left[\begin{smallmatrix} \Omega_{\mathrm{full}}^2 & \epsilon I \\ \epsilon I & I \end{smallmatrix}\right] \succ 0$ with $\epsilon := \gamma/2$. The barrier and the safe set are $h(x) := E_{\mathrm{safe}} - x^{\top}P_{\mathrm{full},\epsilon}x$ and $\mathcal{C} := \{x \in \mathcal{X}_{\mathrm{ODD}} : h(x) \ge 0\}$; under the containment condition (H0) of Theorem 1, $\mathcal{C}$ is the full ellipsoid and lies in the interior of $\mathcal{X}_{\mathrm{ODD}}$ (Lemma S1). The residual F_{nl} may carry a linear part $A_{\mathrm{adv}}x$, its linearisation at the origin; in the benchmark it is the advection, which the certified plant carries through a spectral filter (Section 5).

Reduction and tail. For a full-rank basis $\Phi \in \mathbb{R}^{N\times K}$ let $P_0 := \Phi^{\top}P_{\mathrm{full}}\Phi$, $z(x) := P_0^{-1}\Phi^{\top}P_{\mathrm{full}}x$, $P_{\perp} := I - \Phi P_0^{-1}\Phi^{\top}P_{\mathrm{full}}$, $x_{\perp} := P_{\perp}x$ and $\mathcal{V}_{\perp} := \mathrm{Im}(P_{\perp})$, so that $x = \Phi z + x_{\perp}$. In class (H), $\Phi = \mathrm{blkdiag}(\Phi_q, \Phi_q)$ with Φ_q orthonormal eigenvectors of Ω_{full}^2, which defines $\Omega_r^2 := \Phi_q^{\top}\Omega_{\mathrm{full}}^2\Phi_q$ and $R_{p,\perp} := (I - \Phi_q\Phi_q^{\top})R_p(I - \Phi_q\Phi_q^{\top})$. The reduced coordinates obey exactly

$$\dot{z} = f_r(z) + g_r u + d_r w + \Delta_{\perp}(z, x_{\perp}), \qquad g_r := P_0^{-1}\Phi^{\top}P_{\mathrm{full}}G, \quad d_r := P_0^{-1}\Phi^{\top}P_{\mathrm{full}}D, \tag{1}$$

with $f_r(z) = A_{r,0}z + F_{\mathrm{nl},r}(z)$ the Petrov–Galerkin reduction of F and $\Delta_{\perp}$ the tail coupling.

Assumption 2 (Tail) *(H2a)* $\Phi^\top P_{\mathrm{full},\epsilon} P_\perp = 0$. *(H2b) In class (H), the tail-restricted damping satisfies* $v^\top R_{p,\perp} v > \epsilon \|v\|_2^2$ *for nonzero tail vectors* v. *(H2c) There is* $\lambda_{K+1} > 0$ *with* $-x_\perp^\top (P_{\mathrm{full},\epsilon} A_{\mathrm{full}} + A_{\mathrm{full}}^\top P_{\mathrm{full},\epsilon}) x_\perp \geq 2\lambda_{K+1} \|x_\perp\|_{P_{\mathrm{full},\epsilon}}^2$ *on* $\mathcal{V}_\perp$, *and the tail excitation*

$$M_\perp := \sup_{\mathcal{X}_{\mathrm{ODD}} \times U_{\mathrm{adm}} \times \mathcal{B}(\varepsilon_{\mathrm{adv}})} \|P_\perp [A_{\mathrm{full}} \Phi z(x) + F_{\mathrm{nl}}(x) + Gu + Dw]\|_{P_{\mathrm{full},\epsilon}} < \infty.$$

The tail radius is $\varepsilon_{\mathrm{basis},P} := M_\perp / \lambda_{K+1}$, and the tail coupling constant $L_{\perp,P}$ is the smallest constant with $\|\Delta_\perp(z, x_\perp)\|_2 \leq L_{\perp,P} \|x_\perp\|_{P_{\mathrm{full},\epsilon}}$ for all $x = \Phi z + x_\perp \in \mathcal{X}_{\mathrm{ODD}}$ with $\Phi z \in \mathcal{X}_{\mathrm{ODD}}$. Both are computed from the operators. Under (H2a), $h(\Phi z + x_\perp) = h(\Phi z) - \|x_\perp\|_{P_{\mathrm{full},\epsilon}}^2$ for every $x \in \mathbb{R}^N$, with $h(\Phi z) = E_{\mathrm{safe}} - z^\top P z$ and $P := \Phi^\top P_{\mathrm{full},\epsilon} \Phi$ (Lemma S2).

Controller. The teacher $\hat{\mathcal{T}}(z) := \Pi_U(\mathcal{N}(v))$ acts on the port vector $v := g_r^\top P z \in \mathbb{R}^m$; Π_U is the radial projection onto U_{adm} and $\mathcal{N}$ is an origin-preserving GroupSort-2 network of even hidden width H whose first layer mixes pairwise sum and difference channels at an angle $\theta_{\mathrm{GS}} \in (0, \pi/4)$, whose hidden layers are orthogonal (isometric) and preserve these two channels, and whose output layer is $W_L = -L_{\mathcal{T}}(W_{L-1} \cdots W_1)^\top$ (S17)–(S18). Then $\mathcal{N}$ is $L_{\mathcal{T}}$-Lipschitz and $\hat{\mathcal{T}}$ is $L_{\mathcal{T},z}$-Lipschitz in z with $L_{\mathcal{T},z} := L_{\mathcal{T}} \|g_r^\top P\|_2$. The student $\Psi_B(z) := \Pi_U(\mathcal{N}_q(v))$ uses the b_B-bit fixed-point realization $\mathcal{N}_q$. The look-up table Ψ_A quantizes v by the mid-tread quantizer Q_{b_A} with $2^{b_A} + 1$ levels per axis over the port box of width $D_v := 2 \sup_{\mathcal{C}_{r,0}} \|v\|_\infty$, stores the quantized student's value $\Pi_U(\mathcal{N}_q(v_q))$ at each nonzero grid point v_q and outputs zero on the central bin. With latch delays $0 < \tau_A < \tau_B < T_s$, samples at $t_k = kT_s$, $z_k := z(x(t_k))$ and $v_k := g_r^\top P z_k$, the control is

$$u(t) = \begin{cases} \Psi_B(z_{k-1}), & t \in [t_k, t_k + \tau_A), \\ \Psi_A(Q_{b_A}(v_k)), & t \in [t_k + \tau_A, t_k + \tau_B), \\ \Psi_B(z_k), & t \in [t_k + \tau_B, t_{k+1}), \end{cases} \tag{2}$$

with $z_{-1} := z_0$, so that the register holds $u_{B,-1} := \Psi_B(z_0)$ from standby.

The saturation condition (H-sat) is $L_{\mathcal{T},z} \sqrt{K} D_{\mathrm{ODD}}/2 \leq u_{\max}$ on the reduced box $\mathcal{Z}_{\mathrm{ODD}} := [-D_{\mathrm{ODD}}/2, D_{\mathrm{ODD}}/2]^K$; under it the teacher is sector-aligned on $\mathcal{Z}_{\mathrm{ODD}}$, $-z^\top P g_r \hat{\mathcal{T}}(z) \geq \eta_{\mathcal{T}} \|g_r^\top P z\|_2^2$ with $\eta_{\mathcal{T}} = L_{\mathcal{T}} \cos(2\theta_{\mathrm{GS}}) > 0$ (Lemma S3). With $B_{\max}$ address bits, S_{bits} bits of memory and b_{out} bits per output, the table fits the hardware if $m(b_A + 1) \leq B_{\max}$ and $m(b_A + 1) + \lceil \log_2(m\, b_{\mathrm{out}}) \rceil \leq \log_2 S_{\mathrm{bits}}$. Theorem 1 and (7) apply to any $L_{\mathcal{T},z}$-Lipschitz teacher with certified $\delta_{\mathrm{cert}} > 0$; the GroupSort-2 family parameterised by θ_{GS} is the analytical testbed for which Lemma 1 gives the closed form.

Reduced barrier and shell. All reduced constants are taken over $\mathcal{C}_{r,0} := \{z^\top P z \leq E_{\mathrm{safe}}\}$: $L_h := 2\sqrt{\lambda_{\max}(P) E_{\mathrm{safe}}}$ bounds $\|\nabla h(\Phi z)\|_2$, $\overline{g}_h$ and $\overline{d}_h$ bound $\|2z^\top P g_r\|_2$ and $\|2z^\top P d_r\|_2$, and $M_z := \sup \|f_r(z) + g_r u + d_r w\|_2 + L_{\perp,P} \varepsilon_{\mathrm{basis},P}$ bounds $\|\dot{z}\|_2$. With the dip Δh_{dip} of Section 3, the tightened reduced barrier, its target set, the reduced safe set and the shell partition are

$$\begin{aligned} &\tilde{h}_r(z) := h(\Phi z) - \varepsilon_{\mathrm{basis},P}^2 - \Delta h_{\mathrm{dip}}, \qquad \tilde{\mathcal{C}}_r := \{\tilde{h}_r \geq 0\}, \qquad \mathcal{C}_r := \{h(\Phi z) \geq \varepsilon_{\mathrm{basis},P}^2\}, \\ &\mathcal{C}_{r,\mathrm{act}} := \{z \in \mathcal{C}_r : \tilde{h}_r(z) \leq h_{\mathrm{act}}\}, \qquad \mathcal{B}_{\mathrm{int}} := \mathcal{C}_r \setminus \mathcal{C}_{r,\mathrm{act}}. \end{aligned} \tag{3}$$

The tightening by $\varepsilon_{\mathrm{basis},P}^2$ pays for the tail through Lemma S2, and Δh_{dip} pays for what the barrier may lose between samples. The interior core is the ball $\mathcal{B}_{\mathrm{int}} = \{\|z\|_P < r_{\min}\}$ with $r_{\min}^2 := E_{\mathrm{safe}} - \varepsilon_{\mathrm{basis},P}^2 - \Delta h_{\mathrm{dip}} - h_{\mathrm{act}}$, and the activation level $h_{\mathrm{act}} > L_h M_z T_s$ keeps a trajectory sampled in $\mathcal{B}_{\mathrm{int}}$ out of $\{\tilde{h}_r < 0\}$ for one period, so the budget is needed only on the shell $\mathcal{C}_{r,\mathrm{act}}$.

3. The budget

Along (1), for almost every $t \in [t_k, t_{k+1})$, $\dot{\tilde{h}}_r = \mathcal{L}_{f_r}\tilde{h}_r + \mathcal{L}_{g_r}\tilde{h}_r\, u + \mathcal{L}_{d_r}\tilde{h}_r\, w + \nabla\tilde{h}_r^\top \Delta_\perp$, with $\mathcal{L}_{f_r}\tilde{h}_r(z) = -2z^\top P f_r(z)$, $\mathcal{L}_{g_r}\tilde{h}_r(z) = -2z^\top P g_r$ and $\mathcal{L}_{d_r}h_r(z) = -2z^\top P d_r$. The certified authority δ_{cert} of (6) bounds this rate plus $\gamma_0\tilde{h}_r$ from below at the sample z_k, under the teacher's command $\hat{\mathcal{T}}(z_k)$, with no disturbance and no tail. Each term below bounds one gap between that ideal rate and the plant's.

Tail coupling. Theorem 1(a) keeps $\|x_\perp\|_{P_{\mathrm{full},\epsilon}} \le \varepsilon_{\mathrm{basis},P}$, so $|\nabla\tilde{h}_r^\top\Delta_\perp| \le L_h L_{\perp,P}\varepsilon_{\mathrm{basis},P}$. The constant $L_{\perp,P}$ is at least the norm of the advective block $P_0^{-1}\Phi^\top P_{\mathrm{full}} A_{\mathrm{adv}} P_\perp$, so the tail term is positive whenever this block is nonzero; the rest of $L_{\perp,P}$ comes from the Lipschitz remainder of F_{nl}.

Quantization, verification slack and lag. The student deviates from the teacher by at most $\varepsilon_{\mathrm{CROWN}}(b_B)$, the bound of Proposition 1. The table deviates by at most $\varepsilon_A(b_A) := \varepsilon_{\mathrm{CROWN}}(b_B) + L_{\mathcal{T}}\sqrt{m}D_v/2^{b_A+1}$, the first summand covering table entries computed by the quantized network; the bound also holds on the central bin. All command errors are measured against the teacher at the sample, $\hat{\mathcal{T}}(z_k)$. During $[t_k, t_k+\tau_A)$ the latched command was computed at z_{k-1}; since $\|\dot{v}\|_2 \le M_v := \|g_r^\top P\|_2 M_z$, it differs from $\hat{\mathcal{T}}(z_k)$ by at most $\varepsilon_{\mathrm{CROWN}}(b_B) + \Delta u_{\mathrm{lag}}$ with $\Delta u_{\mathrm{lag}} := L_{\mathcal{T}} M_v T_s = L_{\mathcal{T},z} M_z T_s$. The three latch phases of (2) therefore carry errors $\varepsilon_{\mathrm{CROWN}}(b_B)+\Delta u_{\mathrm{lag}}$, $\varepsilon_A(b_A)$ and $\varepsilon_{\mathrm{CROWN}}(b_B)$, and the exact integration of $\tilde{h}_r$ with decay rate $\gamma_0 > 0$, the linear class-$\mathcal{K}$ gain of the barrier condition in (6), over one period weights them by

$$w_1 := \frac{e^{\gamma_0\tau_A}-1}{e^{\gamma_0 T_s}-1}, \qquad w_2 := \frac{e^{\gamma_0\tau_B}-e^{\gamma_0\tau_A}}{e^{\gamma_0 T_s}-1}, \qquad w_3 := 1-w_1-w_2. \tag{4}$$

The codesign window $\lfloor \log_2(\sqrt{m}D_v/(2M_vT_s))\rfloor + 1 \le b_A \le \lfloor B_{\max}/m\rfloor - 1$ (S30) keeps the table's grid error below Δu_{lag} at its lower end and is the table capacity at its upper end. Proposition 1 is an instance of differential verification (Paulsen et al., 2020; Zhang et al., 2023b) with CROWN-type relaxations (Zhang et al., 2018; Xu et al., 2020) and adds a closed-form companion in the bit width, so that b_B enters the certificate.

Proposition 1 (Twin-network CROWN bound; proof in Proposition S1) *Let $\mathcal{N}_q$ have per-tensor quantized weights with $|[W_\ell^q - W_\ell]_{ij}| \le s_\ell 2^{-b_B}$, so $\|W_\ell^q - W_\ell\|_2 \le \Delta_{W,\ell}$, bias errors of at most $\Delta_{b,\ell}$, and rounding of input, accumulators and activations modelled as injections bounded by ρ_ℓ, with $\Delta_{W,0} = \Delta_{b,0} = 0$. (i) On the shared ReLU routing graph of the GroupSort-2 comparators, every comparator difference of $\Delta_{\mathrm{net}}(z) := \mathcal{N}_q(g_r^\top Pz) - \mathcal{N}(g_r^\top Pz)$ admits a sound two-sided affine relaxation, exact where the comparator sign is fixed in both networks over a box. (ii) Backward substitution over any partition of $\mathcal{Z}_{\mathrm{ODD}}$ into boxes yields $\varepsilon_{\mathrm{CROWN}}(b_B)$ with $\sup_{\mathcal{Z}_{\mathrm{ODD}}}\|\Psi_B - \hat{\mathcal{T}}\|_2 \le \sup_{\mathcal{Z}_{\mathrm{ODD}}}\|\Delta_{\mathrm{net}}\|_2 \le \varepsilon_{\mathrm{CROWN}}(b_B)$. (iii) With $Y := \|g_r^\top P\|_2\sqrt{K}D_{\mathrm{ODD}}/2$ and widths H_ℓ, $\sup_{\mathcal{Z}_{\mathrm{ODD}}}\|\Delta_{\mathrm{net}}\|_2 \le E_q(b_B) := \sum_{\ell=0}^{L}(\Delta_{W,\ell}Y + \Delta_{b,\ell} + \sqrt{H_\ell}\rho_\ell)\prod_{k>\ell}\|W_k^q\|_2$, and the computed ratio $C_{\mathrm{relax}} := \varepsilon_{\mathrm{CROWN}}(b_B)/E_q(b_B)$ is at most $\sqrt{m}$ when every comparator is sign-stable on every box. (iv) If $\Delta_{b,\ell}, \rho_\ell \le c\,2^{-b_B}$, then for fixed architecture, teacher, scales and partition, $E_q(b_B)$ and $\varepsilon_{\mathrm{CROWN}}(b_B)$ are $O(2^{-b_B})$.*

Hold. With $s := t - t_k$, the state has moved by at most $M_z s$, so the certified rate expression $\mathcal{L}_{f_r}\tilde{h}_r + \mathcal{L}_{g_r}\tilde{h}_r u + \gamma_0\tilde{h}_r$ at $z(t)$ differs from its value at z_k by at most $L_{\mathrm{hold}}s$ with $L_{\mathrm{hold}} := (L_{\mathcal{L}_{f_r}\tilde{h}_r} +$

$L_{\mathcal{L}_{g_r}\tilde{h}_r}u_{\max} + \gamma_0 L_h)M_z$. Integrated with the exponential weight, the hold costs $\varphi(\gamma_0 T_s)L_{\text{hold}}T_s$, where $\varphi(\xi) := ((\xi-1)e^{\xi}+1)/(\xi(e^{\xi}-1)) \in (\frac{1}{2}, 1)$. The pointwise deficit that the budget does not absorb inside a period integrates to at most the dip, which (3) reserves,

$$\Delta h_{\text{dip}} := \overline{g}_h(\varepsilon_A(b_A) - \varepsilon_{\text{CROWN}}(b_B))(\tau_B - \tau_A) + \overline{g}_h\, \Delta u_{\text{lag}}\, \tau_A + \tfrac{1}{2}L_{\text{hold}}T_s^2. \tag{5}$$

Adversary. For any admissible $w(t)$, $|\mathcal{L}_{d_r}\tilde{h}_r\, w| \le \overline{d}_h \varepsilon_{\text{adv}}$. The plant and the reduced dynamics (1) are separately affine in u and w, so Isaacs' condition holds for the reduced barrier rate, and the disturbance that minimises that rate pointwise is $w^*_{\text{adv}} = -\varepsilon_{\text{adv}}\, \mathcal{L}_{d_r}\tilde{h}_r^{\top}/\|\mathcal{L}_{d_r}\tilde{h}_r\|_2$ ($w^*_{\text{adv}} = 0$ where $\mathcal{L}_{d_r}\tilde{h}_r = 0$); its Filippov solutions are Carathéodory solutions for some admissible $w(t)$ (Remark S4).

Authority. The certified authority on the shell is

$$\begin{aligned} \delta_{\text{cert}} &:= \inf_{z\in\mathcal{C}_{r,\text{act}}} \left[\mathcal{L}_{f_r}\tilde{h}_r(z) + \mathcal{L}_{g_r}\tilde{h}_r(z)\hat{\mathcal{T}}(z) + \gamma_0\tilde{h}_r(z)\right] \\ &\le\ \delta_{\max} := \inf_{z\in\mathcal{C}_{r,\text{act}}} \sup_{u\in U_{\text{adm}}} \left[\mathcal{L}_{f_r}\tilde{h}_r + \mathcal{L}_{g_r}\tilde{h}_r u + \gamma_0\tilde{h}_r\right]. \end{aligned} \tag{6}$$

On the port null space $\{v = 0\}$ the control term vanishes, so passive dissipation must carry the authority there; the metric is chosen so that it does. The reduced linear part is $A_{r,0} = (J_0 - R_0)P_0$ with J_0 skew, $R_0 \succeq 0$ and $P_0 \succ 0$; write $\mu_{2,P}(A) := \sup_{e\neq 0} e^{\top}(PA + A^{\top}P)e/(2e^{\top}Pe)$. If $R_0 \succ 0$, then $\mu_{2,P_0}(A_{r,0}) \le -\lambda_{\min}(R_0)\lambda_{\min}(P_0) =: -\mu_R$. In class (H), where J_0 is canonical, $P_0 = \text{blkdiag}(\Omega_r^2, I)$ and $R_0 = \text{blkdiag}(0, \gamma I)$, the reduced metric $P_\epsilon := \left[\begin{smallmatrix} \Omega_r^2 & \epsilon I \\ \epsilon I & I \end{smallmatrix}\right]$ with $\epsilon = \gamma/2$ is positive definite, $Q_\epsilon := -(P_\epsilon A_{r,0} + A_{r,0}^{\top}P_\epsilon) = \gamma P_\epsilon$ and $\mu_{2,P_\epsilon}(A_{r,0}) = -\gamma/2 =: -\mu_R$ (Proposition S2, which also gives the admissible range of ϵ).

Lemma 1 (Class-(H) authority, $m \le K$; proof in Lemma S5) *Let (H0)–(H3) hold, let the reduced model be in class (H) with $R_0 = \text{blkdiag}(0, \gamma I)$, $P = P_\epsilon$, $\epsilon = \gamma/2$, and let the shell be fixed with $r_{\min}^2 > 0$. Split $F_{\text{nl}}(x) = A_{\text{adv}}x + n(x)$ with $A_{\text{adv},r} := P_0^{-1}\Phi^{\top}P_{\text{full}}A_{\text{adv}}\Phi$, $n_r(z) := P_0^{-1}\Phi^{\top}P_{\text{full}}n(\Phi z)$ and $\|P^{1/2}n_r(z)\|_2 \le s_{\text{nl}}\|z\|_P$ on $\mathcal{C}_{r,\text{act}}$. With $M := \gamma P - (PA_{\text{adv},r} + A_{\text{adv},r}^{\top}P) + 2\eta_{\mathcal{T}}Pg_rg_r^{\top}P$ and $c_M := \lambda_{\min}(M, P) - 2s_{\text{nl}}$,*

$$\delta_{\text{cert}} \ge \min\left\{c_M r_{\min}^2 + \gamma_0 h_{\text{act}},\ \ c_M(r_{\min}^2 + h_{\text{act}} + \Delta h_{\text{dip}}) - \gamma_0 \Delta h_{\text{dip}}\right\},$$

the minimum being the first branch exactly when $c_M \ge \gamma_0$.

Collecting the terms gives the budget, the Certified Control Authority Condition (H6):

$$\begin{aligned} \delta_{\text{total}}(K, b_A, b_B, T_s) := \ & \underbrace{L_h L_{\perp,P}\varepsilon_{\text{basis},P}}_{\text{tail}} + \underbrace{\overline{g}_h w_2(\varepsilon_A(b_A) - \varepsilon_{\text{CROWN}}(b_B))}_{\text{table}} + \underbrace{\overline{g}_h\varepsilon_{\text{CROWN}}(b_B)}_{\text{network}} \\ & + \underbrace{w_1\overline{g}_h\,\Delta u_{\text{lag}}}_{\text{lag}} + \underbrace{\varphi(\gamma_0 T_s)L_{\text{hold}}T_s}_{\text{hold}} + \underbrace{\overline{d}_h\varepsilon_{\text{adv}}}_{\text{adversary}} \ \le\ \delta_{\text{cert}}, \end{aligned} \tag{7}$$

which is (S33) with the table and network parts separated.

4. Main result

Theorem 1 (Full-order safety under multi-rate quantized control) *Assume (H0)* $E_{\mathrm{safe}} < \lambda_{\min}(P_{\mathrm{full},\epsilon})\rho_{\mathrm{ODD}}^2$ *and* $E_{\mathrm{safe}} < \frac{1}{4}D_{\mathrm{ODD}}^2\lambda_{\min}(P)$*; (H1) Assumption 1, with* $L_{F_{\mathrm{nl}}}$, $M_\perp$, $L_{\perp,P}$ *taken over* $\mathcal{X}_{\mathrm{ODD}}$*; (H2) Assumption 2 ((H2a)–(H2c)), and in class (H),* $\gamma < 2\omega_{\min}$ *and* $\epsilon = \gamma/2$*; (H3) the teacher construction of Section 2* (S17)–(S18) *with* $H/2 \geq m$, $G_{\mathrm{port}} := g_r^\top P^2 g_r \succ 0$*, and (H-sat); (H4) the shell condition* $\varepsilon_{\mathrm{basis},P}^2 + \Delta h_{\mathrm{dip}} + L_h M_z T_s < E_{\mathrm{safe}}$ *and* $h_{\mathrm{act}} \in (L_h M_z T_s,\ E_{\mathrm{safe}} - \varepsilon_{\mathrm{basis},P}^2 - \Delta h_{\mathrm{dip}})$ *(equivalently* $r_{\min}^2 > 0$*); (H5)* $b_A \geq 1$ *satisfies the table capacity conditions of Section 2 (the lower end of the codesign window of Section 3 is optional); (H6) the Certified Control Authority Condition (7),* $\delta_{\mathrm{total}}(K, b_A, b_B, T_s) \leq \delta_{\mathrm{cert}}$ $(\leq \delta_{\max})$*, with* δ_{cert} *defined by (6); (H7)* $x(0) = \Phi z(0) + x_\perp(0)$ *with* $z(0) \in \tilde{\mathcal{C}}_r$ *and* $\|x_\perp(0)\|_{P_{\mathrm{full},\epsilon}} \leq M_\perp/\lambda_{K+1}$*. Then the Carathéodory solution* $x(t)$ *of the plant under the multi-rate law (2) and any measurable* $w(t) \in \mathcal{B}(\varepsilon_{\mathrm{adv}})$ *exists on* $[0,\infty)$ *and satisfies, for all* $t \geq 0$ *and* $k \in \mathbb{N}_0$*: (a)* $\|x_\perp(t)\|_{P_{\mathrm{full},\epsilon}} \leq \varepsilon_{\mathrm{basis},P}$*; (b)* $\tilde{h}_r(z(t_k)) \geq 0$ *and* $\tilde{h}_r(z(t)) \geq -\Delta h_{\mathrm{dip}}$*, so* $z(t) \in \mathcal{C}_r$*; (c)* $h(x(t)) \geq \tilde{h}_r(z(t)) + \Delta h_{\mathrm{dip}} \geq 0$*, so* $x(t) \in \mathcal{C}$.

Proof sketch (full proof in the supplement, proof of Theorem S1). Step 1 (existence of the Carathéodory solution and reduction to the first exit of x from $\mathcal{X}_{\mathrm{ODD}}$ or of z from $\mathcal{C}_{r,0}$) and Step 5 (Lemma S2 and (a) split $h(x)$ into the reduced barrier and the tail, which gives (c)) are standard. Step 2, tail invariance: (H2a) makes the $P_{\mathrm{full},\epsilon}$-norm of the tail obey a scalar inequality with no cross term; (H2c) and $M_\perp$ give $\frac{d}{dt}\frac{1}{2}\|x_\perp\|_{P_{\mathrm{full},\epsilon}}^2 \leq -\lambda_{K+1}\|x_\perp\|_{P_{\mathrm{full},\epsilon}}^2 + M_\perp\|x_\perp\|_{P_{\mathrm{full},\epsilon}}$, which with (H7) yields (a). Step 3, exact integration over one period: if $z_k \in \mathcal{B}_{\mathrm{int}}$, Lipschitz continuity of $\tilde{h}_r$ and $h_{\mathrm{act}} > L_h M_z T_s$ keep $\tilde{h}_r > 0$ over the period. If $z_k \in \mathcal{C}_{r,\mathrm{act}}$, adding and subtracting (6) at z_k bounds $\dot{\tilde{h}}_r + \gamma_0 \tilde{h}_r$ below by δ_{cert} minus the tail and adversary terms, the shift $L_{\mathrm{hold}} s$ and $\overline{g}_h$ times the command error of the current phase. Integrating against $e^{\gamma_0 s}$ over the three latch phases of (2) gives exactly the weights (4), the factor φ and the six terms of (7), so (H6) yields $\tilde{h}_r(z_{k+1}) \geq 0$. Step 4, the dip: within a period the instantaneous deficit is bounded phase by phase, and its integral is at most Δh_{dip} of (5), which gives (b).

Stochastic extension. Corollary S1 of the supplement adds a globally Lipschitz diffusion to the plant and lets the disturbance law drift at Wasserstein-2 rate r_{W_2} over a horizon T; the tail radius is then replaced by an Ornstein–Uhlenbeck envelope $\varepsilon_{\mathrm{basis},\sigma}$, and $\varepsilon_{\mathrm{adv}}$ by $\varepsilon_{\mathrm{adv}} + r_{W_2} T$ in all constants, including δ_{cert}. The Itô correction carries the budget into a stochastic budget $\delta_{\mathrm{total},\sigma} + \delta_{\mathrm{stoch}}(\alpha) + \delta_{\mathrm{drift}} \leq \delta_{\mathrm{cert}}$, in which δ_{stoch} contains $\gamma_0 h_{\mathrm{act}}$. If it closes under the hypotheses of Theorem 1 and the remaining conditions of that corollary, and $T \leq T_{\mathrm{TTL}}$, the smallest of a drift-reserve bound, a Kushner-type bound (Kushner, 1967) and a tail horizon, then $\mathbb{P}(\inf_{t\in[0,T]} h(x(t)) \geq 0) \geq 1 - \alpha$. The experiments use $r_{W_2} = 0$, and the corollary is feasible in regime R1 and infeasible in R2 (Section 5).

5. Numerical certification

Benchmark. The plant is a class-(H) telegraph reaction–diffusion–advection lattice, $\dot{q} = p$, $\dot{p} = -(\nu L_\Delta + \alpha_0 I)q - \gamma p - c W_f D_h W_f q - \kappa(q - \sin q) + Bu + B_w w$ on 160 nodes (320 states), with L_Δ the discrete Laplacian, $\nu = 0.05$, $\alpha_0 = 1$, $\gamma = 1$, $\kappa = 0.05$, $m = 2$ actuators and one disturbance channel with $\varepsilon_{\mathrm{adv}} = 0.002$. The skew central difference D_h is filtered by $W_f = \Phi_f\,\mathrm{diag}(1,1,w,w)\Phi_f^\top$ on the first 4 sine modes; the deployed cutoff-mode weight is $w^\star =$

Table 1: The six terms of the budget (7), the authority of Lemma 1 against its brute-force minimum and the utilisation (supplement, Experiments 1 and 3); the budget closes in both regimes.

	R1 (Péclet 1)		R2 (Péclet 20)	
term	value	share	value	share
tail, $L_h L_{\perp,P} \varepsilon_{\mathrm{basis},P}$	1.835×10^{-5}	0.006967	0.003666	0.2704
table (LUT), $\overline{g}_h w_2(\varepsilon_A(b_A) - \varepsilon_{\mathrm{CROWN}}(b_B))$	7.392×10^{-6}	0.002806	7.393×10^{-5}	0.005452
network (CROWN), $\overline{g}_h \varepsilon_{\mathrm{CROWN}}(b_B)$	2.172×10^{-4}	0.08246	0.002133	0.1573
lag, $w_1 \overline{g}_h \Delta u_{\mathrm{lag}}$	2.26×10^{-5}	0.008579	1.859×10^{-4}	0.01371
hold, $\varphi(\gamma_0 T_s) L_{\mathrm{hold}} T_s$	0.001474	0.5596	0.006607	0.4872
adversary, $\overline{d}_h \varepsilon_{\mathrm{adv}}$	8.944×10^{-4}	0.3396	8.944×10^{-4}	0.06596
δ_{total}	0.002634		0.01356	
δ_{cert}, lower bound (Lemma 1)	0.03161		0.01456	
δ_{cert}, brute force	0.03164		0.01468	
utilisation $\delta_{\mathrm{total}}/\delta_{\mathrm{cert}}$	0.0833		0.931	

$2^{-11/2} \approx 0.0221$, selected by the sweep below. The advection stays in F_{nl}, so A_{full} is the pure damped wave, $Q_{\gamma/2} = \gamma P_{\gamma/2}$ holds to 7.039×10^{-15}, and (H2a) holds by eigenmode invariance. The reduced model keeps $K_m = 2$ modes ($K = 4$), fewer than the advection couples, so the tail term is active. The teacher is constructed, not trained, with $L = 3$ layers, $H = 8$ and $\theta_{\mathrm{GS}} = 0.05$; the student's rounding offsets are learned, which lowers $\varepsilon_{\mathrm{CROWN}}(b_B)$ at $b_B = 12$ from 0.004718 (nearest rounding) to 0.004455.

Regime R2 has $c = 1$ (Péclet 20) and open-loop growth rate 0.2925, and 16 of 16 open-loop seeds leave $\mathcal{C}$; regime R1 has $c = 0.05$ (Péclet 1). Both use $E_{\mathrm{safe}} = 0.05$, $B_{\max} = 24$, $\log_2 S_{\mathrm{bits}} = 26$, $b_{\mathrm{out}} = 16$ and $b_B = 12$; R2 (R1) uses $\gamma_0 = 0.35$ (0.3), $u_{\max} = 3.4$ (0.35), $L_{\mathcal{T}} = 5$ (0.5), $T_s = 3 \times 10^{-4}$ (0.002), $\tau_A = 1.5 \times 10^{-5}$ (1×10^{-4}) and $\tau_B = 3 \times 10^{-5}$ (2×10^{-4}). The table width $b_A = 9$ lies in the codesign window $[8, 11]$, and the certificate refuses unless (H0)–(H7) all hold, with (H5) checked on this window.

Soundness of the parts. Over 100016 samples of $\mathcal{Z}_{\mathrm{ODD}}$ and b_B from 3 to 14, the twin-network bound of Proposition 1 exceeds the largest sampled student–teacher gap by a factor of 1.887 to 2.844 and roughly halves with every added bit. The class-(H) bound of Lemma 1 is within one percent of the brute-force minimum in both regimes (Table 1); in all numbers below, δ_{cert} denotes this certified lower bound. Over 16 seeds per regime under the rate-minimising adversary w^*_{adv}, from boundary starts (tail at its largest admissible value) and interior starts, conclusions (a)–(c) of Theorem 1 hold at every integrator node.

Conservatism by term. At the 102636 integrator nodes of the certified R2 runs whose sample lies in $\mathcal{C}_{r,\mathrm{act}}$, each pointwise term of the rate inequality behind (7) is compared with the bound the proof uses: the adversary term reaches 0.8772 of its bound, quantization and lag 0.3729, tail coupling 0.3161 and the hold term 0.07439. The hold term, the largest share of δ_{total} in R2 (Table 1), is the least realised, through $\|z - z_k\|_2$ (at most 0.3205 (R2) and 0.5853 (R1) of $M_z s$) and the Lipschitz constant of the reduced rate (at most 0.2323 and 0.1794). The realised rate exceeds the proof's lower bound by at least 0.6597 δ_{total} (R2) and 11.5 δ_{total} (R1) at every shell node, which suggests that a trajectory-dependent bound in place of M_z and these constants could lengthen the admissible period (medians: Experiment 4b of the supplement).

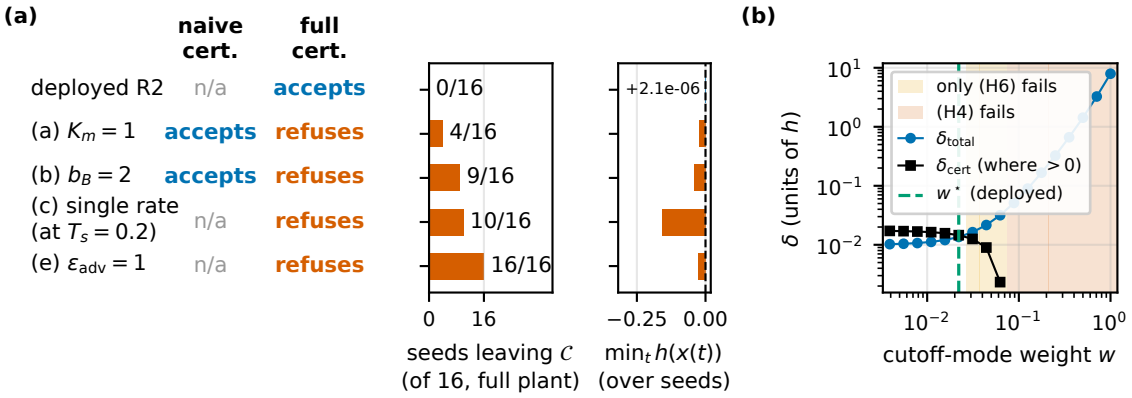


Figure 1: Ablations and the frontier in the certified loop. (a) Naive and full verdicts, the number of the 16 seeds whose full-order trajectory leaves the safe set, and $\min_t h(x(t))$ over the seeds, for the deployed R2 configuration (Experiment 4) and ablations (a), (b), (c) and (e) (Experiment 6, horizon $T = 3$ instead of 5). Row (c) is refused at the deployed period, where none of its seeds exits; its exits are at $T_s = 0.2$, where both loops are uncertified. Where the naive certificate accepts and the full certificate refuses, the plant leaves the safe set. (b) δ_{total} and δ_{cert} (where positive) against the cutoff-mode weight w at Péclet 20, with the bands where only (H6) fails and where (H4) fails; the certificate accepts up to $w^\star$ and refuses above it.

Ablations. Each ablation alters one component at the deployed $w^\star$, and the full certificate refuses each (Figure 1(a)). Ablation (a) is a projection-based reduced-order certificate without a tail term, at $K_m = 1$: the naive certificate accepts, and 4 of 16 seeds leave the safe set. Ablation (b) is a 2-bit network certified as if it were its teacher: the naive certificate without the quantization term accepts, and 9 of 16 seeds leave the safe set. Ablation (c) holds a single command per period; at the deployed period it has $\delta_{\text{total}} = 0.01702 > \delta_{\text{cert}} = 0.01456$ and is refused, while the multi-rate loop has $\delta_{\text{total}} = 0.01356$, and none of its seeds leaves the safe set there within the horizon. At $T_s = 0.2$, where both loops are uncertified, the single-rate loop exits on 10 of 16 seeds; the multi-rate loop at that period is reported in the supplement. Ablation (e) raises the adversary to $\varepsilon_{\text{adv}} = 1$: all 16 seeds exit, and at the largest certifiable level $\varepsilon^\star_{\text{adv}} = 0.004223$ none does.

Refusals and the frontier. The certificate refuses 6 configurations: a look-up table indexed by z instead of v, whose codesign window is empty, its lower end exceeding its upper end $(9, 5)$; $u_{\max}$ below the (H-sat) bound; unfiltered advection; the plain projector $w = 1$; the longer period $T_s = 5 \times 10^{-4}$, where $\delta_{\text{total}} = 0.01821 > \delta_{\text{cert}} = 0.01455$; and all timing constants scaled by 10. Over 17 values of w from 1 down to 0.003906, with the open loop unsafe at every value, the certificate accepts for $w \leq 0.0221$ and refuses above it, through (H6) and then (H4) (Figure 1(b)). For the plain projector, on the scanned grid, the certified advection speeds ($c \leq 0.02$, Péclet at most 0.4) and the open-loop-unstable ones ($c \geq 0.5$, Péclet at least 10) are disjoint. The frontier is set by the supremum over the L^2-type domain $\mathcal{X}_{\text{ODD}}$ in $M_\perp$, which charges cutoff-mode advection at full strength. The deadzone result of Proposition S3 holds in R1, where $\mu_R = 0.5$ exceeds

the P-norm sector modulus $L_{\mathrm{sec}} = 0.1204$ of $F_{\mathrm{nl},r}$: the mid-tread deadzone silences the table late in the run, while the mid-rise quantizer of ablation (d) chatters at every tested b_A (between 121.6 and 953.1 sign switches per unit time; Experiment 5 of the supplement); in R2, $L_{\mathrm{sec}} = 2.392 > \mu_R$ and it does not apply. Corollary S1 is exercised in R1 at noise level $\sigma^\star = 0.007$: 0 of 2000 paths exit, the Clopper–Pearson upper limit 0.001843 lies below $\alpha = 0.05$, and the horizon bound $T_{\mathrm{TTL}} = 2022$ exceeds the horizon $T = 2$ by a factor 1011. The corollary is infeasible in R2 because the stochastic budget adds $\gamma_0 h_{\mathrm{act}} = 0.007237$ to δ_{total}, more than the margin 9.994×10^{-4}. ETD2RK step-halving ratios are 4.002 to 4.045. Every constant is computed in float64 from the operators, and the repository's 248 assertions and 65 tests regenerate every number (code: `github.com/AnuKarthikaS/pgcbf`).

6. Limitations and conclusion

The certified plant is the lattice with filtered advection at $w^\star = 0.0221$; unfiltered advection and the plain projector are refused, and certifying full-strength cutoff-mode advection at the unstable Péclet numbers would need a different formulation, for example an H^1-bounded domain, whose effect on a smooth field the filter weight mimics heuristically (Remark S2); the theorem does not have one. The sampling period $T_s = 3 \times 10^{-4}$ is set by the certificate, not by the plant, which is sampled 1.714×10^4 times per slowest plant period in R2; the hold term is 0.4872 of δ_{total}, the price of a continuous-time guarantee with a Lipschitz hold bound (Section 5). On every certified run the minimum of h is attained at the initial instant, and the dip floor $-\Delta h_{\mathrm{dip}}$ is never reached. Theorem 1 assumes exact acquisition of z_k and exact latch instants; rounding of the Path-A input by ε_z adds $L_{\mathcal{T},z}\varepsilon_z$ to $\varepsilon_A(b_A)$ (Remark S3).

The stochastic extension is feasible in R1 only, its horizon bound is 1011 times the tested horizon, and every stochastic run uses $r_{W_2} = 0$. Proposition S3 does not apply in R2 because $L_{\mathrm{sec}} > \mu_R$ there. The evidence is one benchmark class in two regimes. Ablations (a), (b) and (c) are paradigm-level baselines, not reimplementations of published methods; (b) is a 2-bit network and (c) has no naive verdict. Below $w^\star$ the single-rate budget closes (first at $w = 0.01562$), so $w^\star$ is the largest grid value at which every ablation is refused. Under the quadratic barrier, $\theta_{\mathrm{GS}} = 0$ (linear port feedback) maximises $\eta_{\mathcal{T}}$, so on this benchmark the neural teacher is not needed for authority; it is the object whose quantization and verification the budget must charge. A task on which a nonlinear neural policy beats linear port feedback on δ_{cert} is future work. The margin in R2 is $\delta_{\mathrm{cert}} - \delta_{\mathrm{total}} = 9.994 \times 10^{-4}$.

Each of the six gaps between the verified model and the running plant costs one term of (7), and Theorem 1 turns a closed budget into full-order safety for all time; on the lattice the budget closes at utilisation 0.931 and names the hold term as the one to sharpen. The budget is the certificate, and its refusals are results.

Generative AI Use

During the preparation of this manuscript, Anthropic's Claude Opus 5 and GitHub Copilot were used strictly for prose copy-editing, boilerplate code and unit-test generation, and BibTeX formatting. All mathematical formulations, theoretical derivations, proofs, core implementations, and scientific claims are solely the work of the author, who takes full responsibility for the content.

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